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1. Challenge Title: 3D industrial object detection: Field Device Detector

2. Problem Statement What is the business or technical problem to solve? Context, why it matters, and pain points:

Industrial Metaverse is based on 3D Digital Twin from the mill/factory. Each mill has thousands of field devices in different places. They are tank level measurements, pressure or flow measurements. As a service engineer walks on the site humans can see & detect those easily in one second, so should AI also find those devices.

3. Guiding Question Focus the challenge: How can we use AI to detect those devices?

4. Recommended AI Approaches

There are several existing object detection ML algorithms like AutoML, SSD, Yolo, SAM v2 and latest new ones are VLM (Vision Language Models). Interesting possibilities that should be trained and tested and demonstrated.

5. Dataset or API Provided

Valmet will provide 3D Gaussian splatting file (.ply) from the pilot plant and video that contains field devices.

Small 200 photos of the different field devices that can be used to train models will be provided.

6. Evaluation Criteria How should solutions be judged? What's most important - e.g., accuracy, scalability, usability, business impact?

Detection, accuracy and speed are important. Trained model could be part of the remote support / training tool in Valmet´s future Industrial Metaverse product.

7. Submission Format What deliverables do you expect? (e.g., demo video, pitch slides, code repo, documentation)

Demo video and presentation, code with the model (to be agreed later).

8. Point of Contact (if remote) Name, email, and preferred contact method for questions during hackathon

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